

Hyunwon Chung



INTERESTS

Efficient AI, LLM Approximation, Wireless Communication, Machine Learning, Reconfigurable architecture, CGRA, VLIW Processor, Dataflow Architecture, Ultra Low-Power SoC

EDUCATION

University of Michigan, Ann Arbor

MS, PhD. in Electrical and Computer Engineering

Aug 2022 – Current

Korea University, Seoul

BS. in Electrical Engineering (Include 2-year military service)

Mar 2015 – Feb 2022

RESEARCH

Ultra Low-power Audio Compression Processor

Co-advisors: Prof. David Blaauw and Prof. Hun-Seok Kim

- Developed performance and power models for compressed neural network architectures targeting real-time audio compression acceleration
- Modeled energy and performance trade-offs across convolution operators by analyzing loop ordering, dataflow strategies, and on-chip buffer/SRAM sizing
- Performed architecture-level design space exploration of systolic arrays, evaluating array dimensions and microarchitectural variants to identify hardware configurations optimized for compressed models

PROWESS (Processor Reconfiguration for Wideband Sensor Systems)

Co-advisors: Prof. David Blaauw and Prof. Hun-Seok Kim

- Developed a streaming processor with a hardware kernel launcher enabling 10 ns runtime reconfiguration without CPU intervention
- Designed a dynamic scheduling unit for graph-based, data-dependent kernel execution under multi-channel RF workloads
- Taped out the processor in 28 nm CMOS and delivered up to 8.3× end-to-end performance improvement and 5× PE utilization via concurrent execution and speculative preloading
- Achieved 248.5 GFLOPS/mm² compute density and 1.67 TFLOPS/W energy efficiency in 28 nm CMOS

COCHON (Configurable Optical Communications via Heterogeneous-Processing Optimized Node)

Co-advisors: Prof. David Blaauw and Prof. Hun-Seok Kim

- Contributed to a 130 mm² heterogeneous processor integrating optical transceiver and receiver digital baseband for beyond-5G communication systems
- Led top-level physical design and multi-clock SoC integration, driving robust clock-domain crossing (CDC) architecture and interface timing closure across multiple asynchronous accelerator domains
- Successfully taped out end-to-end silicon in the GF 12 nm process, coordinating cross-team integration across accelerator, baseband, and SoC subsystems

DASH-SoC (Domain-Focused Advanced Software Reconfiguration Heterogeneous)

Co-advisors: Prof. David Blaauw and Prof. Hun-Seok Kim

- Contributed to the development of two accelerators under the DARPA-funded DASH project
- DAP (Domain Adaptive Processor)
 - Redesigned the DAP compute architecture from 16-bit to 32-bit datapath, significantly improving compute capability and performance
 - Implemented multi-kernel execution support to efficiently handle diverse communication workloads
 - Led physical design optimization, resolving routing congestion and improving timing closure and silicon efficiency
- Fully Reconfigurable Accelerator for FEC

- Architected a flexible forward-error-correction processor supporting 3GPP LTE/5G, WiFi, and SDA OCT standards
- Integrated six FEC modes within a unified architecture, enabling high protocol adaptability with minimal area overhead

Automatic Peak Detection Algorithm based on Continuous Wavelet Transform

Advisors: Prof. Yogesh Gianchandani

University of Michigan

- Developed software and algorithm pipeline for multi-detector micro-GC chromatogram analysis
- Improved peak-detection and baseline-correction accuracy using CWT-based sparse feature extraction
- Co-authored publication accepted to Journal of Chromatography A (2024)

Low Power Accelerators for ML & Communication Systems

Advisor: Prof. Jongsun Park

Korea University

- Designed a DCT-based JPEG encoder/decoder architecture and synthesized the RTL in 65nm TSMC Technology
- Developed a low-power hardware implementation of a Viterbi decoder architecture
- Proposed an 8-bit fixed-point low-power accelerator for convolutional neural networks
- Designed a folded architecture for fully-connected layers to improve compute efficiency

PUBLICATION

Hyunwon Chung, Parin Senta, Jason Yu, Yukun Fang, Pierre Abillama, Kuan-Yu Chen, Ronald Dreslinski, David Blaauw, Hun-Seok Kim. "A 248.5 GFLOPS/mm², 1.67 TFLOPS/W Streaming Processor with Hardware-Level Scheduling for Advanced Spectrum Sensing." IEEE Symposium of VLSI Circuits (2026)

Sanghyuck Moon, Ashfakh Hluvallay, **Hyunwon Chung**, Myeongsu Ko, Seokhyeon Jeong, Jungho Lee, Mohammad Khreishah, Jeongtaek Chang, Hung Do, Caitlyn Sutherland, Jason Kapit, David Nicholson, William Reinhardt, Dennis Sylvester, Mark Miskin, David Blaauw. "A 95.0 dB Dynamic Range Zero-Bias PV Light-to-Digital Converter for Seawater Monitoring with Single Point Calibration." IEEE Symposium of VLSI Circuits (2026)

Yufan Yue, Kuan-Yu Chen, Xiangdoing Wei, Tutu Ajayi, **Hyunwon Chung**, Ronald Dreslinski, David Blaauw, Hun-Seok Kim. "QFEC: A 9.97Gb/s Fully Configurable Quad-Mode Decoder for LDPC, Polar, Turbo, and Convolutional Codes." IEEE Transactions on Circuits and Systems I: Regular Papers (2026)

Jiahao Lin, H. Umut Suluhan, **Hyunwon Chung**, , Arindam Dutta, Anish Vipperla, Gerard Gubash, Jacob Holtom, Bernd-Peter Paris, Chaitali Chakrabarti, Daniel W. Bliss, David Blaauw, Hun-Seok Kim, Ali Akoglu, Umit Y. Orgas. "An Overview of Challenges and Requirements for Real-Time Spectrum Sensing in Modern RF Autonomy Systems." IEEE Design & Test (2025)

Xiangyu Zhao, Ryan Aridi, Jacob Hume, Swetha Subbiah, Xingqi Wu, **Hyunwon Chung**, Yutao Qin, and Yogesh B. Gianchandani. "Automatic peak detection algorithm based on continuous wavelet transform for complex chromatograms from multi-detector micro-scale gas chromatographs." Journal of Chromatography A (2024)

SELECTED PROJECTS

A Dynamic Quantized CNN Processor with Analog MAC Operations

- *EECS 627 course project at the University of Michigan*
- Designed analog MAC unit which includes digital to pulse converter, analog datapath, and flash ADC
- Proposed dynamic quantization method which allows each layer have different fixed points
- Post-quantization training and quantization-aware training are both used for 5-bit quantization, to get proper accuracy on MNIST and Cifar-10 dataset

16 bit Microprocessor with In-memory computing technology based on 6T SRAM architecture

- *EECS 427 course project at the University of Michigan*
- Developed 16-bit Microprocessor based on RISC-V ISA, which includes RF, ALU, Controller, Program Counter, Shifter, and IMEM/DMEM
- Designed customized 6T-SRAM based in-memory computing (IMC) module into 16-bit microprocessor

Custom Design Projects

- Implemented 128-bit Lightweight Encryption Algorithm (LEA) Decryption security module using Matlab & Verilog and synthesized it using 90nm TSMC technology

- Developed several circuit design including Inverter Chain, Ring Oscillator, SRAM, 3-tap FIR filter, and Flip-Flops. This includes the schematic, layout, and post-layout simulation

The World Embedded Software Contest - AI Humanoid Division

- Developed a program for real-time image processing and humanoid robot control using python
- Won 3rd place and Awarded Embedded SW & System Industry Association Chairman's award

Computer Architecture Project

- Developed Inverse-matrix program and heap-sort program using ARM assembly language
- Implemented TCP Server-Client program using multi-thread technique in C

SKILLS

Languages : C, C++, Verilog, SystemVerilog, Python, OpenMP, CUDA, PyTorch, Tensorflow
 Hardware Simulation Tool : PSPICE, Design Compiler, Model Sim, VCS
 Circuit Design Software : Virtuoso, Innovus, Calibre

MILITARY SERVICE

Korean Augmentation To the United States Army,
Squad Leader, Sergeant, Headquarter, 23rd CBERN, 2ID, Camp Humphreys Sep 2016 – Jun 2018

AWARDS & SCHOLARSHIPS

Electrical and Computer Engineering Department Graduate Fellowship, University of Michigan	2025
Dean's list at School of Electrical Engineering, Korea University	2018 - 2021
Miraero Scholarship at School of Electrical Engineering, Korea University	2016
National Scholarship (Admission with highest distinction), Korea University	2015